

1. Proof by induction

Lets prove that:

$$(1 + 2 + \dots + n)^2 > 1^2 + 2^2 + \dots + n^2 \text{ for } n \geq 2$$

Let's rewrite both of the sums of n numbers:

- sum of n numbers: $1 + 2 + \dots + n = \left(\frac{n(n+1)}{2}\right)^2$
- square sum of n numbers: $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Now we can try to prove that $\left(\frac{n(n+1)}{2}\right)^2 > \frac{n(n+1)(2n+1)}{6}$ and simplify it further

1.1. Base case: $n = 2$

$$\left(\frac{2(2+1)}{2}\right)^2 = 3^2 = 9 > \frac{2(2+1)(4+1)}{6} = \frac{30}{6} = 5$$

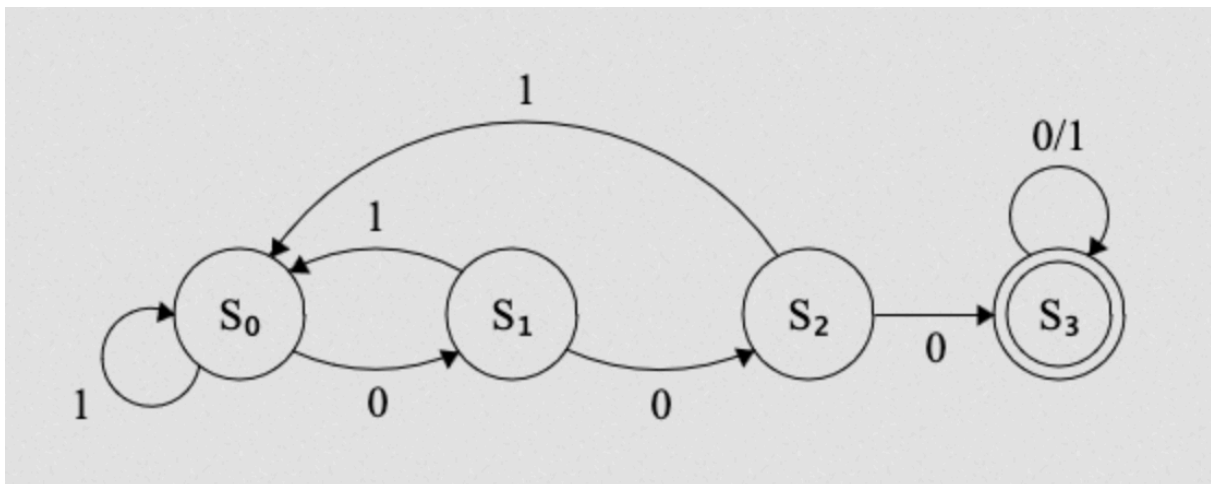
1.2. Inductive step:

The prove holds for the base case. Now we can assume that the prove holds for k and try to prove it for $k + 1$.

$$\begin{aligned} \left(\frac{(k+1)(k+2)}{2}\right)^2 &> \frac{(k+1)(k+2)(2(k+1)+1)}{6} \\ \left(\frac{k^2+3k+2}{2}\right)^2 &> \frac{2k^3+9k^2+13k+6}{6} \\ \frac{k^4+6k^3+13k^2+12k+4}{4} &> \frac{2k^3+9k^2+13k+6}{6} \end{aligned}$$

Since we know that $k \geq 2$ we can say that $k^4 \geq 4k^3$ and $k^2 \geq 2k$ and $12k \geq 13$ and $4 \geq 6$.

2. Triple 0 state machine



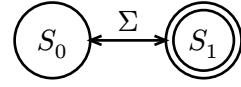
Here we have a simple state machine that only allows through itself binary strings that include a substring of "000".

3. Is B a regular language

Suppose that A is a regular language with alphabet Σ . Let B denote the language where we have omitted strings with an even length

$$B = \{w \in A : |w| = 2k + 1 \text{ for some } k = 1, 2, \dots\}$$

We can show that B is **regular** by creating a state machine for the language. We can only go between S_0 and S_1 , and we can only end in S_1 if we have an odd number of letters in the input word.



4. Calculating checksum

4.1. Proofing that $w^R \in A$ for all $w \in A$

We know that

$$c = a_0 \oplus a_1 \oplus \dots \oplus a_n$$

and w^R , being the reverse of w , should correspond to

$$c^R = a_n \oplus a_{n-1} \oplus \dots \oplus a_0$$

Because of the commutitive properties of XOR we know that $a_0 \oplus a_1 = a_1 \oplus a_0$ and since c^R is just c reversed we can safely say that $c = c^R$ and therefore we proof that $w^R \in A$ for all $w \in A$